



Adyapan

School

Genomics



Duration - 2 months

**Industry
Certification**



Skill India Certified

**250+
Partner Companies**

From **genes** to insights

Master modern **genomics.**

From genetic data to biological insights - become industry-ready.

This immersive Genomics program takes you beyond foundational concepts into real-world genomic analysis workflows. Learn DNA sequencing technologies, variant analysis, genome annotation, and data interpretation through hands-on projects and guided pipelines. With a strong focus on practical implementation and research-driven applications, you'll graduate with the ability to analyze large-scale genomic datasets, extract meaningful biological insights, and contribute to advancements in healthcare, biotechnology, and precision medicine.

8

WEEKS

30+

PROGRAMS OFFERED

20,000+

STUDENTS

250+ PARTNERED COMPANIES



ABOUT ADYAPAN SCHOOLS

Where education meets real-world impact

Not just a course — a platform to launch
your career.

Adyapan Schools was built with a single conviction:
learning works best when it happens in the real world.
We partner with top companies, mentors, and industry
platforms to ensure every student graduates with a
portfolio of work that speaks louder than a certificate.

Our programs combine rigorous coursework with live
client projects, giving you the skills and proof-of-work
that employers actually want.

MISSION

To equip ambitious learners with
practitioner-level digital
marketing skills through mentor-
led, project-based education that
bridges the gap between learning
and earning.



VISION

To be India's most trusted
launchpad for the next generation
of marketing leaders — defined
not by degrees but by the real
work.



Everything you need to grow fast

PROGRAM HIGHLIGHTS



Live Industry Projects

Work on campaigns for real brands alongside your coursework. Build portfolio projects that prove your expertise to employers.



1-on-1 Mentorship

Dedicated mentors from Google, Microsoft, Mastercard and more. Get personalized guidance and industry connections.



AI-Powered Marketing

Learn cutting-edge AI tools alongside evergreen fundamentals. Stay ahead of the curve in a rapidly evolving landscape.



Dual Certification

Earn both a Course Completion and Internship Certificate – accredited by Skill India Digital Hub and NSDC.



Internship Guarantee

Graduate with an internship completion certificate from a live brand project. Concrete, resume-ready proof of work.



Industry Network

Join a network of alumni at Amazon, Google, Adobe, Microsoft. Access exclusive hiring events and referral opportunities.

8 weeks. 8 modules. Infinite impact.

WEEK 1

Foundations of Genomics & Molecular Biology

- Discussion of curriculum
- Overview of the central dogma
- Understand genome organisation
- Study the human genome
- Introduction to genome databases
- Understand different types of genetic variation: SNPs, indels, CNVs, and structural variants
- Explore the concept of genotype vs phenotype and how variation drives disease



WEEK 2

DNA Sequencing Technologies & Platforms

- Understand first-generation sequencing
- Study Next-Generation Sequencing (NGS): Illumina sequencing by synthesis, paired-end reads, and flow cells
- Introduction to Third-Generation Sequencing
- Compare sequencing platforms by read length, accuracy, throughput, and cost per base
- Understand sequencing library preparation: fragmentation, adapter ligation, and PCR amplification
- Explore key sequencing applications



WEEK 3

Bioinformatics Foundations & Sequence Analysis

- Set up your bioinformatics environment
- Understand FASTQ format, Phred quality scores, and raw read quality assessment using FastQC
- Perform read trimming and adapter removal using Trimmomatic and Cutadapt
- Study reference genome alignment
- Understand SAM/BAM file formats and post-alignment processing with SAMtools and Picard
- Introduction to genome assembly concepts



8 weeks. 8 modules. Infinite impact.

WEEK 4

Variant Calling & Annotation

- Understand the GATK Best Practices pipeline for germline and somatic variant calling
- Perform BQSR and duplicate marking before variant calling
- Call SNPs and indels using GATK HaplotypeCaller and Mutect2 for somatic variants
- Understand VCF file format
- Annotate variants using ANNOVAR, SnpEff, and VEP (Variant Effect Predictor) for functional impact
- Filter and prioritise variants using population frequency databases



WEEK 5

Transcriptomics & RNA-Seq Analysis

- Understand the transcriptome and how gene expression differs across tissues, time points, and conditions
- Perform RNA-Seq alignment using STAR and pseudo-alignment
- Quantify gene expression
- Perform differential gene expression (DGE) analysis using DESeq2 and edgeR in R
- Conduct Gene Ontology (GO) enrichment and KEGG pathway analysis using clusterProfiler
- Introduction to alternative splicing analysis and visualisation



WEEK 6

Epigenomics & Functional Genomics

- Understand epigenetic mechanisms
- Study ChIP-Seq for mapping protein-DNA interactions and histone marks
- Introduction to ATAC-Seq for profiling open chromatin and identifying active regulatory elements
- Analyse DNA methylation
- Explore enhancer-promoter interactions and topologically associating domains (TADs)
- Use ENCODE and Roadmap Epigenomics databases to contextualise functional genomic findings



8 weeks. 8 modules. Infinite impact.

WEEK 7

Single-Cell Genomics & Multi-Omics Integration

- Understand the principles of single-cell RNA sequencing (scRNA-Seq)
- Process scRNA-Seq data using the Seurat pipeline: quality control, normalisation, and dimensionality reduction
- Perform cell clustering, UMAP visualisation, and automated cell type annotation using SingleR
- Study trajectory inference and pseudotime analysis using Monocle3 to reconstruct developmental paths
- Introduction to multi-omics integration: combining genomics, transcriptomics, proteomics, and metabolomics



WEEK 8

Genomics in Medicine, Ethics & Career Readiness

- Understand clinical genomics applications: rare disease diagnosis, pharmacogenomics, and cancer genomics
- Study genome-wide association studies (GWAS)
- Explore polygenic risk scores (PRS) and their application in personalised medicine and disease risk prediction
- Introduction to CRISPR-based genomic editing and its intersection with functional genomics
- Understand ethical, legal, and social issues (ELSI) in genomics



WHO THIS IS FOR

This course is **perfect** for

Students & Career Switchers

Aspiring Genomics &
Bioinformatics Professionals

Biotechnology, Genetics &
Molecular Biology Students

Data Analysts Transitioning
to Biological Data

Healthcare, Precision Medicine &
Drug Discovery Enthusiasts

Researchers & Lab
Professionals

CERTIFICATIONS



ALUMNI NETWORK

Our alumni work at world-class companies

Amazon

Adobe

Google

Autodesk

Microsoft

Deloitte

Your career switch is one click away.

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us at support@adyapan.com

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